

Joseph Barbosa

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EDUCATION

University of Waterloo (Double Degree with Wilfrid Laurier University) GPA: 3.96 / 4.0 (92%)
Bachelor of Computer Science (AI Specialization, Stats Minor) & Bachelor of Business Administration
Awards: President's Gold Scholarship (\$21,000), Loblaw National Scholarship (\$2,000), Term Distinction (All Terms)

EXPERIENCE

Undergraduate Researcher Sep. 2026 – Present
University of Waterloo *Waterloo, ON*

- Researching **multi-agent** LLM systems for heterogeneous healthcare data sensemaking and clinical question decomposition.

Software Engineering Intern May 2026 – Aug. 2026
Scotiabank *Toronto, ON*

- Spearheaded Flowpilot: a VLM-powered ReAct QA agent using Set-of-Marks grounding for live UI test automation. Achieved a **78%** one-shot pass rate on a held-out **33-case** evaluation suite. Reduced QA time from **2h to 20m** across **5 services**.
- Designed AI architecture for ProfileAI (shipping FY27, **50K clients/day**): vector search, pre-filtering, and prompt caching pipelines; collaborated with internal and MongoDB engineers to design production-scale inference and retrieval strategies.

Software Engineering Intern Jan. 2026 – Apr. 2026
ESGTree *Waterloo, ON*

- Built a production NL-to-SQL agent over a **98-table** MySQL database using ReAct reasoning, schema retrieval, & join-aware prompting; achieved **90.9%** accuracy on **77** eval queries, crediting self-correction (**+13.0pp**) & RAG (**+7.8pp**) gains.
- Engineered production backend with SQL validation, read-only DB access, and multi-tenant isolation via JWT; designed a 3-model cost/latency tier and Docker-sandboxed Python execution for data analytics through a Next.js interface.

Case Competitor & Team Delegate 2025 – Present
Wilfrid Laurier University *Waterloo, ON*

- Won **\$3,000+** across case competitions; selected as delegate for Laurier's digital strategy and international case teams.
- Designed an AI-assisted team workflow (mockups, transcriptions, and solutioning), adopted by international teammates.

Software Development Intern Jun. 2024 – Dec. 2024
FuturIQ *Brampton, ON*

- Built an end-to-end ML pipeline for housing price prediction: automated Selenium data ingestion of **1,000+** current listings and engineered **8** features, deploying the regularized polynomial model via Flask/React with realtime automated retraining.

PROJECTS

Studeal | *Python, FastAPI, Playwright, Celery, Redis, Postgres + pgvector, AWS, Next.js* | [GitHub](https://github.com) studeal.site

- Built an AI deal-finder using concurrent agents to search **10** marketplaces via AX-tree perception, achieving **97.7%** precision on a held-out marketplace with zero site-specific code. Moved per-page extraction from **13 to 106** via overlapping chunks.
- Engineered a conversational agent using a per-turn JSON state machine to elicit and record buyer intent, then embedded it as a multimodal preference vector for product recommendation via cosine similarity, LLM reranking, & hard prefiltering.
- Shipped live product hunt streaming, a shared listing pool, billing, and email on AWS via Terraform and Bedrock.

Scholr | *Python, FastAPI, Next.js, sentence-transformers, OpenAI* | [GitHub](https://github.com) tryscholr.com

- Built a **multi-agent** research assistant over **200M+ papers** by decomposing user queries into subtopics researched by parallel agents leveraging recursive, bounded-depth planning. Deployed via **MCP** and a **Next.js** application to **300+ users**.
- Engineered a two-stage retrieve-and-rerank system (bi-encoder and cross-encoder), lifting **nDCG@5 by 10.8%** over an ablated, single-stage dense (bi-encoder only) retrieval on an LLM-judged benchmark of **20** handwritten LLM evaluation queries.
- Hardened for production with deterministic citation validation guaranteeing **100%** of cited sources resolve to real papers.

PantryPal | *Python, PyTorch, QLoRA, DPO, XGBoost* | [GitHub](https://github.com) [Blog Post](#)

- Investigated whether weak domain classifiers could replace human preference labels for LLM alignment; preprocessed **10K** RecipeNLG examples to train **18** XGBoost models across dietary categories, generating **1,200+** DPO preference pairs.
- Fine-tuned Llama 3.2 3B with QLoRA SFT on **8K** recipes, reaching **99.7% keyword compliance** vs. **56.8%** GPT-4o.

TECHNICAL SKILLS

Languages: Python, SQL, TypeScript, JavaScript, C, C++

AI / ML: PyTorch, Hugging Face Transformers, PEFT, LangGraph, MCP, fine-tuning, RAG, AI evals, OpenAI & Anthropic APIs

Frameworks: React, Next.js, FastAPI, Flask, Node.js

Libraries: pandas, NumPy, scikit-learn, XGBoost, Pydantic, SQLAlchemy, Selenium, Playwright

Databases / Cloud: PostgreSQL, Redis, pgvector, Azure, AWS

Tools: Git, Docker, Bash, Claude Code, CI/CD, GitHub Actions

Interests: Marathon running, strength training, road cycling, golfing, cooking